

Market Efficiency and Inefficiency

The efficient markets hypothesis, return predictability, anomalies, multifactor models, and the limits of arbitrage

Tobias J. Moskowitz

Dean Takahashi Professor of Finance, Yale School of Management

MGT 595 · Quantitative Investing

Lecture 3

What this deck covers

- **Defining market efficiency**

Prices are correct, no free lunch — and what “correct” presumes

- **The three forms and their tests**

Weak, semi-strong, strong; the Grossman–Stiglitz paradox; the joint hypothesis problem

- **Random walks and martingales**

Fair games, the random walk hypotheses, autocorrelation tests, cross-autocorrelations

- **Market anomalies**

The recipe, and the three rival readings: risk, inefficiency, or data mining

- **Multifactor models of risk and return**

Good and bad times beyond the market; consumption-based intuition

- **Finding the factors**

Statistical factors, macro factors, characteristic portfolios (Fama–French)

- **Introduction to anomalies**

Size, January and seasonals, contrarian reversals, value and growth, momentum

- **The efficiency debate and limits to arbitrage**

Shleifer's three arguments, Shleifer–Vishny limits

- **Case study: can the market add and subtract?**

Lamont–Thaler (2003): 3Com/Palm, negative stubs, and short-sale constraints

Defining Market Efficiency

Market efficiency means ...

Five statements that all say the same thing.

Prices are correct: they fully reflect all available information

$$P_t = \mathbb{E}_t \left(\frac{d_{t+1}}{1 + r_{t+1}} + \frac{d_{t+2}}{(1 + r_{t+2})^2} + \dots \right)$$

People use **all available information** in forming expectations about future cash flows.

The **discount rate is right** for the riskiness of the cash flows.

Prices react to *new* information **quickly** and to the **right extent**.

There is **no free lunch**: the only way to earn higher returns is to take on more risk — no information out there can be used to construct strategies that earn more than required for their risk.

The hidden assumption

When we say “prices are correct,” we are implicitly stating what *correct* is — i.e. we are **assuming an asset pricing model**.

What information are we talking about?

Three nested information sets define three forms of efficiency.

Information in **past prices**

Public information

Private information

We can characterize markets as

weak | **semi-strong** | **strong**

-form efficient if all

past price | public | private

information is incorporated in prices already.

What information are we talking about?

Three nested information sets define three forms of efficiency.

Information in **past prices**

Public information

Private information

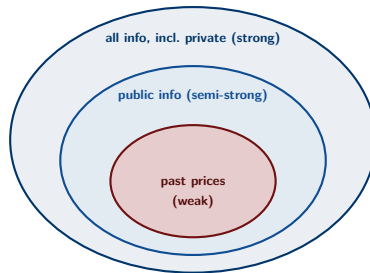
We can characterize markets as

weak | **semi-strong** | **strong**

-form efficient if all

past price | public | private

information is incorporated in prices already.



A paradox, and its resolution

Paradox

If markets are efficient, then it does **not pay to search for information**. But how, then, can markets be efficient if no one searches for information and incorporates it into prices?

A paradox, and its resolution

Paradox

If markets are efficient, then it does **not pay to search for information**. But how, then, can markets be efficient if no one searches for information and incorporates it into prices?

Transactions and trading costs play the key role

Prices reflect information up to the point where the **marginal benefit** of acting on information no longer exceeds the **marginal cost** of acquiring and trading on it.

A paradox, and its resolution

Paradox

If markets are efficient, then it does **not pay to search for information**. But how, then, can markets be efficient if no one searches for information and incorporates it into prices?

Transactions and trading costs play the key role

Prices reflect information up to the point where the **marginal benefit** of acting on information no longer exceeds the **marginal cost** of acquiring and trading on it.

So someone has to do the searching

Some investors **will** — and **should** — search for information, and will profit from it.

Who are these investors?

A paradox, and its resolution

Paradox

If markets are efficient, then it does **not pay to search for information**. But how, then, can markets be efficient if no one searches for information and incorporates it into prices?

Transactions and trading costs play the key role

Prices reflect information up to the point where the **marginal benefit** of acting on information no longer exceeds the **marginal cost** of acquiring and trading on it.

So someone has to do the searching

Some investors **will** — and **should** — search for information, and will profit from it.

Who are these investors?

Operational definition of efficiency

We need to determine whether deviations from *extreme* market efficiency are **within information and trading costs**.

How can we tell if markets are *inefficient*?

Look for strategies based on past information that earned high returns with little risk.

Unfortunately, we can **never be sure** of inefficiency:

How can we tell if markets are *inefficient*?

Look for strategies based on past information that earned high returns with little risk.

Unfortunately, we can **never be sure** of inefficiency:

it is always possible that we are *not measuring risk properly*,

i.e., we don't know what the right discount rate is.

How can we tell if markets are *inefficient*?

Look for strategies based on past information that earned high returns with little risk.

Unfortunately, we can **never be sure** of inefficiency:

it is always possible that we are *not measuring risk properly*,

i.e., we don't know what the right discount rate is.

The Joint Hypothesis Problem

Market efficiency itself is **not testable**. It must be tested *jointly* with a model of prices / expected returns. A rejection always has two readings: the market is inefficient, *or* the model of expected returns is wrong.

How can we tell if markets are *inefficient*?

Look for strategies based on past information that earned high returns with little risk.

Unfortunately, we can **never be sure** of inefficiency:

it is always possible that we are *not measuring risk properly*,

i.e., we don't know what the right discount rate is.

The Joint Hypothesis Problem

Market efficiency itself is **not testable**. It must be tested *jointly* with a model of prices / expected returns. A rejection always has two readings: the market is inefficient, *or* the model of expected returns is wrong.

Why would we expect markets to be efficient?

The forces of arbitrage — smart investors exploit the mispricing in securities until it disappears.

The Three Forms and Their Tests

1. Weak form efficiency

Weak form efficient

Current prices *fully* reflect all information in **past prices**.

Fama (1991) denotes the slightly broader heading of “**past return predictability**” in general — e.g. including D/P , E/P , size, BE/ME, ... predictability.

⇒ ‘**technical**’ analysis using past price patterns will *not* produce profits.

Examples of weak-form questions

Do prices **over-** or **under-react**?

Are there **seasonal patterns**?

Prices *should* take this into account — but do they?

Are deviations within transactions and trading costs?

2. Semi-strong form efficiency: event studies

Semi-strong form efficient

Current prices fully reflect all past prices **and all publicly available information**.

⇒ **'fundamental' analysis** (e.g. sorting through income statements) will not produce profits.

Public news announcements will be incorporated into prices **quickly** → no profitable trading rule.

Event study analysis: e.g. announcement of a takeover — look at the stock price reaction to the news, *averaged over many companies*.

Are deviations within transactions and trading costs?

The price runs up *before* the announcement (information leakage / anticipation), jumps on the news, and is **flat afterward** — consistent with rapid incorporation of public information.

2. Semi-strong form efficiency: event studies

Semi-strong form efficient

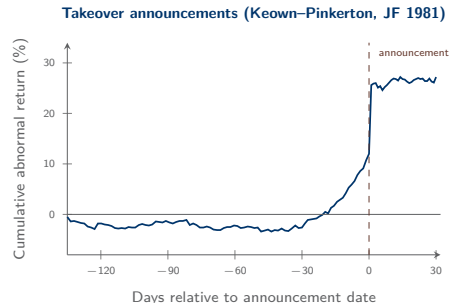
Current prices fully reflect all past prices **and all publicly available information**.

⇒ **'fundamental' analysis** (e.g. sorting through income statements) will not produce profits.

Public news announcements will be incorporated into prices **quickly** → no profitable trading rule.

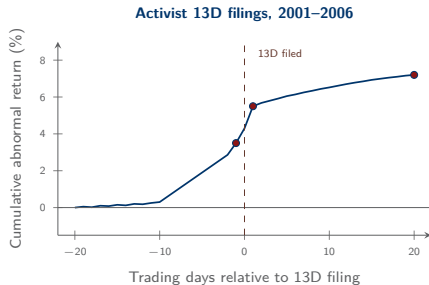
Event study analysis: e.g. announcement of a takeover — look at the stock price reaction to the news, *averaged over many companies*.

Are deviations within transactions and trading costs?



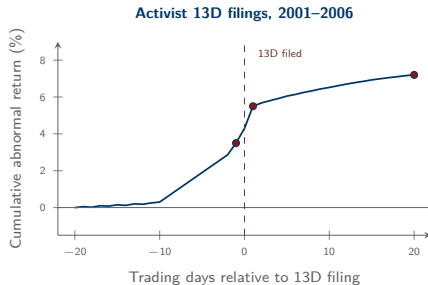
The price runs up *before* the announcement (information leakage / anticipation), jumps on the news, and is **flat afterward** — consistent with rapid incorporation of public information.

A more recent event: activist hedge funds file a 13D



Source: Brav, Jiang, Partnoy and Thomas, "Hedge Fund Activism, Corporate Governance, and Firm Performance," *Journal of Finance* 63(4), 2008, 1729–1775. Path plotted from the CARs reported in that paper.

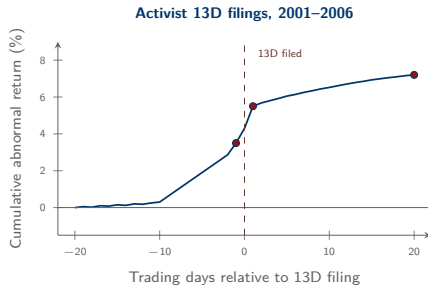
A more recent event: activist hedge funds file a 13D



- Run-up of $\approx 3.2\%$ from day -10 to -1 : the market moves *before* the public filing.

Source: Brav, Jiang, Partnoy and Thomas, "Hedge Fund Activism, Corporate Governance, and Firm Performance," *Journal of Finance* 63(4), 2008, 1729–1775. Path plotted from the CARs reported in that paper.

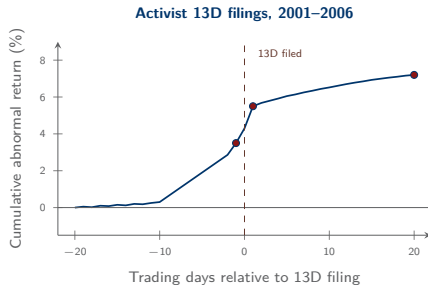
A more recent event: activist hedge funds file a 13D



- Run-up of $\approx 3.2\%$ from day -10 to -1 : the market moves *before* the public filing.
- Jump of $\approx 2.0\%$ on the filing day and the day after.

Source: Brav, Jiang, Partnoy and Thomas, "Hedge Fund Activism, Corporate Governance, and Firm Performance," *Journal of Finance* 63(4), 2008, 1729–1775. Path plotted from the CARs reported in that paper.

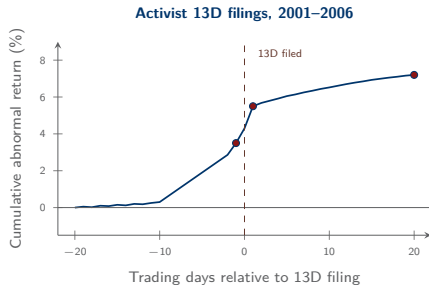
A more recent event: activist hedge funds file a 13D



- Run-up of $\approx 3.2\%$ from day -10 to -1 : the market moves *before* the public filing.
- Jump of $\approx 2.0\%$ on the filing day and the day after.
- Then the CAR **keeps drifting up**, to $\approx 7.2\%$ by day $+20$ — *not* the flat line of Keown–Pinkerton.

Source: Brav, Jiang, Partnoy and Thomas, "Hedge Fund Activism, Corporate Governance, and Firm Performance," *Journal of Finance* 63(4), 2008, 1729–1775. Path plotted from the CARs reported in that paper.

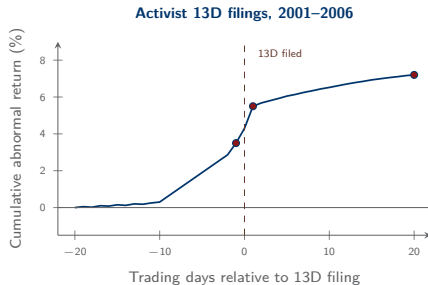
A more recent event: activist hedge funds file a 13D



- **Run-up of $\approx 3.2\%$** from day -10 to -1 : the market moves *before* the public filing.
- **Jump of $\approx 2.0\%$** on the filing day and the day after.
- Then the CAR **keeps drifting up**, to $\approx 7.2\%$ by day $+20$ — *not* the flat line of Keown–Pinkerton.
- And it is **not reversed**: no negative drift over the following year.

Source: Brav, Jiang, Partnoy and Thomas, "Hedge Fund Activism, Corporate Governance, and Firm Performance," *Journal of Finance* 63(4), 2008, 1729–1775. Path plotted from the CARs reported in that paper.

A more recent event: activist hedge funds file a 13D



- Run-up of $\approx 3.2\%$ from day -10 to -1 : the market moves *before* the public filing.
- Jump of $\approx 2.0\%$ on the filing day and the day after.
- Then the CAR **keeps drifting up**, to $\approx 7.2\%$ by day $+20$ — *not* the flat line of Keown–Pinkerton.
- And it is **not reversed**: no negative drift over the following year.

So is this efficient?

A 20-day drift after a *public* filing is hard to square with semi-strong efficiency — unless the 13D merely starts a process whose news arrives gradually. **The joint hypothesis problem, again.**

Source: Brav, Jiang, Partnoy and Thomas, "Hedge Fund Activism, Corporate Governance, and Firm Performance," *Journal of Finance* 63(4), 2008, 1729–1775. Path plotted from the CARs reported in that paper.

3. Strong form efficiency

Strong form efficient

Current prices fully reflect *all* information, **public and private**.

⇒ even **insider trading** will not produce profits.

Example: knowing a merger is going to take place *before* it is announced publicly.

The hierarchy

Strong form ⇒ semi-strong form ⇒ weak form. Each form nests the previous information set, so rejecting a weaker form rejects all stronger forms.

Random Walks and Martingales

Martingales: the original “fair game”

It is often thought that efficient markets $\Rightarrow$ prices move randomly. Let's be precise.

The martingale hypothesis

A martingale is a stochastic process $\tilde{P}_t$ which satisfies

$$\begin{aligned}\mathbb{E}[\tilde{P}_{t+1} \mid P_t, P_{t-1}, \dots] &= P_t \\ \mathbb{E}[\tilde{P}_{t+1} - P_t \mid P_t, P_{t-1}, \dots] &= 0\end{aligned}$$

Can be thought of as a **fair game**, which does not favor one side or the other.

The best estimate of tomorrow's price is **today's price**.

This implies price changes (returns) should **not be serially correlated** at all leads and lags — returns should not be predictable!

But risk and return should be related

Since the martingale hypothesis does *not account for risk*, it is **neither necessary nor sufficient** for rationally determined prices in an efficient market. If there is drift in the process, returns *will* be predictable — but the drift could be due to risk.

The random walk hypothesis

A closely related but stronger model.

RW1: iid increments

Consider the simplest case, where price changes are iid:

$$P_t = \mu + P_{t-1} + \epsilon_t, \quad \epsilon_t \sim \text{iid}(0, \sigma^2)$$

where μ is the expected price change, or *drift*. Then

$$\mathbb{E}[P_t | P_0] = \mu t + P_0, \quad \text{Var}[P_t | P_0] = \sigma^2 t$$

— the conditional mean and variance are **linear in time**.

The random walk is also a fair game, but in a **stronger sense** than the martingale: *independence* implies not only that increments are uncorrelated, but that **any nonlinear functions** of the increments are uncorrelated too.

Weaker random walk hypotheses

RW2: independent, but not identically distributed

Allows for **heteroskedasticity** and non-stationary price changes (more realistic).

Still: any arbitrary transformation of future price increments is unforecastable using any arbitrary transformation of past price increments.

RW3: uncorrelated, but not independent, increments

The **weakest** RW hypothesis of all — contains the other two as special cases.

Example: $\text{Cov}(\epsilon_t, \epsilon_{t-k}) = 0$, but $\text{Cov}(\epsilon_t^2, \epsilon_{t-k}^2) \neq 0$ (volatility clustering).

When does efficiency imply a random walk?

If the discount rate for an asset *does not change* over time, then it **is** true that efficient markets $\Rightarrow$ random walk — e.g. over **short time frames**, returns should look random. This means predictability tests run head-on into the **joint hypothesis problem**; researchers usually employ short-term returns to mitigate it.

Testing the RW: the autocorrelation coefficient

The most direct and intuitive test: serial correlation in the time series of an asset's return.

Autoregression at lag k and the autocorrelation coefficient

$$\begin{aligned}\tilde{r}_{it} &= \gamma_{0,t} + \gamma_{1,k} r_{i,t-k} + \tilde{\epsilon}_{it} \\ \gamma_{1,k} &= \frac{\text{Cov}(r_{it}, r_{i,t-k})}{\sigma^2(r_{i,t-k})} \\ \rho(r_{it}, r_{i,t-k}) &= \frac{\text{Cov}(r_{it}, r_{i,t-k})}{\sigma(r_{it}) \sigma(r_{i,t-k})} \\ \rho^2(r_{it}, r_{i,t-k}) &= \frac{\gamma_{1,k}^2 \sigma^2(r_{i,t-k})}{\sigma^2(r_{it})} = \text{proportion of variance attributed to } r_{i,t-k}\end{aligned}$$

Under iid returns

$$\sigma(r_{it}) = \sigma(r_{i,t-k}) = \sigma(r_i) \quad \forall k \quad \implies \quad \rho(r_{it}, r_{i,t-k}) = \gamma_{1,k}$$

Autocorrelation = autoregression coefficient for lag k .

A striking empirical pattern

Individual stocks

At short horizons (daily, weekly, monthly), $\gamma_{1,k}$ is **negative**.

Portfolios of stocks

At the same short horizons, $\gamma_{1,k}$ is **positive**.

How can both be true?

There must be large positive **cross-autocorrelations** across individual stocks over time — some stocks' past returns predict *other* stocks' future returns.

The next slide decomposes the portfolio autocovariance to see exactly where the positive sign comes from.

Decomposing the portfolio autocovariance

Consider the equal-weighted market portfolio $R_{mt} = \sum_{i=1}^n \frac{1}{n} r_{it}$.

Own- vs cross-autocovariances

$$\begin{aligned}\text{Cov}(R_{mt}, R_{m,t-k}) &= \text{Cov}\left(\sum_{i=1}^n \frac{1}{n} r_{it}, \sum_{i=1}^n \frac{1}{n} r_{i,t-k}\right) \\&= \left(\frac{1}{n}\right)^2 \sum_{i=1}^n \text{Cov}(r_{it}, r_{i,t-k}) + \sum_{i=1}^n \left(\frac{1}{n}\right)^2 \sum_{j \neq i}^n \text{Cov}(r_{jt}, r_{i,t-k}) \\&= \frac{1}{n^2} \sum_{i=1}^n \text{Cov}(r_{it}, r_{i,t-k}) + \frac{1}{n} \sum_{i=1}^n \frac{1}{n} \sum_{j \neq i}^n \text{Cov}(r_{jt}, r_{i,t-k}) \\&= \text{own autocovariances} + \text{cross-autocovariances}\end{aligned}$$

Reading the decomposition

The own terms are small and negative (individual stocks); the portfolio autocovariance is positive, so the **cross terms must be large and positive** — e.g. large stocks lead small stocks, and **liquid stocks lead illiquid stocks**.

Does short-term predictability mean markets are inefficient?

No: not necessarily

It could be driven *spuriously* by **microstructure effects** (bid–ask bounce, nonsynchronous trading).

Evidence

Predictable components are a **small** part of the variance of daily, weekly, or monthly returns.

But they explain as much as **40% of the variance of long-horizon returns** (2–10 years).

Long-term returns are **negatively autocorrelated** over 3–5 year horizons (Fama and French 1988a).

Does short-term predictability mean markets are inefficient?

No: not necessarily

It could be driven *spuriously* by **microstructure effects** (bid–ask bounce, nonsynchronous trading).

Evidence

Predictable components are a **small** part of the variance of daily, weekly, or monthly returns.

But they explain as much as **40% of the variance of long-horizon returns** (2–10 years).

Long-term returns are **negatively autocorrelated** over 3–5 year horizons (Fama and French 1988a).

Two caveats on the long-horizon evidence

Small-sample issues mean these tests have **low power** (not so reliable).

Could be consistent with *irrational bubbles* **as well as** *mean-reverting expected returns* (rational) — the joint hypothesis problem again.

Market Anomalies

The market anomalies

What is an anomaly?

Investment strategies which seem to earn **high returns without being very risky** — they improve the Sharpe ratio of the mean–variance frontier above the market portfolio's.

The recipe

- (1) Form a portfolio based on **observable characteristics**, and measure its returns over time.
- (2) Does the strategy give **high returns on average**?
- (3) Don't forget **transactions costs**: frictions to trading and exploiting the anomaly (bid–ask spread, transactions costs, liquidity, taxes, etc.).

If the answer is yes: three rival readings

1. Risk

The strategy may be **risky**, and the high average returns are just fair compensation for that risk. But — how do we measure risk?

2. Inefficiency?

If risk does not explain the high returns, is it evidence of market inefficiency? **No** — not yet:

It may be *spurious*, the result of **data mining**: try many strategies and some will do great in historical data — telling you nothing about future performance.

Or simply **poor risk measurement**.

Even if no risk measure turns up ...

The risk may be present but just *hasn't surfaced* within the sample analyzed (peso problem), or we're just *not looking at the right measure* of risk.

Multifactor Models of Risk and Return

General ideas in asset pricing

A security's expected return depends on when it makes its payments.

Pays up in bad times

That's exactly when you *need* the money!

⇒ the security is attractive

⇒ you are prepared to accept a **low** expected return.

Pays up in good times

You don't need any more money at these times.

⇒ the security is unattractive

⇒ you require a **high** expected return.

In the CAPM

Good and bad times are captured by **the market return** — good times: market return is high, and vice-versa. Securities with high β pay up precisely when the market is doing well — what use is that? ⇒ they must earn a high expected return. This is exactly what the CAPM predicts.

But there is more to good and bad times than the market return

GNP growth, recession vs. boom, changes in the inflation rate, ... Perhaps it matters how assets pay off *relative to these variables*. Hence researchers began to explore **multifactor models** such as the **APT** and the **ICAPM**.

Consumption-based and intertemporal models

Where do the “factors” come from in theory?

Investors face **consumption and savings decisions** and do not consume all of their wealth period by period.

They live for multiple periods and thus face a **dynamic portfolio decision**: how much to save, how much to consume, and what portfolio to hold.

They trade off the *marginal-utility loss* of consuming less today (to buy more of an asset) against the *marginal-utility gain* of the asset's future payoff. Price is set where **marginal loss equals marginal gain**.

The pricing principle

The asset's price is the **expected discounted value of its payoff, using marginal utility to discount**. Risk corrections are determined by the **covariance of payoffs with consumption**.

Implication

Assets that do badly when consumption is *low* (recessions) are **less desirable (more risky)** than assets that do badly when consumption is *high* (booms).

Finding the Factors

Two theories behind multiple factors: APT and ICAPM

The Arbitrage Pricing Theory (Ross, 1976)

Assume returns have a **factor structure**: a few common factors drive the covariance among assets,

$$\tilde{r}_i = \mathbb{E}[r_i] + b_{i,1}\tilde{F}_1 + \cdots + b_{i,K}\tilde{F}_K + \tilde{\epsilon}_i, \quad \mathbb{E}[\tilde{F}_k] = 0, \quad \tilde{\epsilon}_i \text{ idiosyncratic.}$$

Diversification kills the $\tilde{\epsilon}_i$; **no arbitrage** then forces expected returns to be linear in the loadings:

$$\mathbb{E}[r_i] = r_f + b_{i,1}\lambda_1 + \cdots + b_{i,K}\lambda_K.$$

The factors are whatever drives **common variation**; the theory is *silent on their identity*.

Two theories behind multiple factors: APT and ICAPM

The Arbitrage Pricing Theory (Ross, 1976)

Assume returns have a **factor structure**: a few common factors drive the covariance among assets,

$$\tilde{r}_i = \mathbb{E}[r_i] + b_{i,1}\tilde{F}_1 + \cdots + b_{i,K}\tilde{F}_K + \tilde{\epsilon}_i, \quad \mathbb{E}[\tilde{F}_k] = 0, \quad \tilde{\epsilon}_i \text{ idiosyncratic.}$$

Diversification kills the $\tilde{\epsilon}_i$; **no arbitrage** then forces expected returns to be linear in the loadings:

$$\mathbb{E}[r_i] = r_f + b_{i,1}\lambda_1 + \cdots + b_{i,K}\lambda_K.$$

The factors are whatever drives **common variation**; the theory is *silent on their identity*.

The Intertemporal CAPM (Merton, 1973)

Long-lived investors care about market risk and about **shifts in the investment opportunity set**, summarized by state variables $S_1, \dots, S_K$. They *hedge* those shifts, so expected returns carry premia for the hedging betas:

$$\mathbb{E}[r_i] - r_f = \beta_{i,M}\lambda_M + \beta_{i,S_1}\lambda_{S_1} + \cdots + \beta_{i,S_K}\lambda_{S_K}.$$

Here theory *does* name the factors: **state variables** that forecast changing investment opportunities or consumption.

Two theories behind multiple factors: APT and ICAPM

The Arbitrage Pricing Theory (Ross, 1976)

Assume returns have a **factor structure**: a few common factors drive the covariance among assets,

$$\tilde{r}_i = \mathbb{E}[r_i] + b_{i,1}\tilde{F}_1 + \cdots + b_{i,K}\tilde{F}_K + \tilde{\epsilon}_i, \quad \mathbb{E}[\tilde{F}_k] = 0, \quad \tilde{\epsilon}_i \text{ idiosyncratic.}$$

Diversification kills the $\tilde{\epsilon}_i$; **no arbitrage** then forces expected returns to be linear in the loadings:

$$\mathbb{E}[r_i] = r_f + b_{i,1}\lambda_1 + \cdots + b_{i,K}\lambda_K.$$

The factors are whatever drives **common variation**; the theory is *silent on their identity*.

The Intertemporal CAPM (Merton, 1973)

Long-lived investors care about market risk and about **shifts in the investment opportunity set**, summarized by state variables $S_1, \dots, S_K$. They *hedge* those shifts, so expected returns carry premia for the hedging betas:

$$\mathbb{E}[r_i] - r_f = \beta_{i,M}\lambda_M + \beta_{i,S_1}\lambda_{S_1} + \cdots + \beta_{i,S_K}\lambda_{S_K}.$$

Here theory *does* name the factors: **state variables** that forecast changing investment opportunities or consumption.

Why this matters for what follows

Statistical factors (Method 1) are **APT**-consistent; macro variables (Method 2) and characteristic portfolios (Method 3) appeal to the **ICAPM**. Keep asking: does the motivation fit the method?

We need to find some factors

There are three ways to find factors.

1. Use a **statistical procedure** — factor analysis or principal components — to design factor portfolios.
2. Use **macroeconomic variables** like changes in interest rates or inflation.
3. Use **firm characteristics** to design portfolios to track the factors.

What we will ask of each

Its *advantage*, its *disadvantage*, the classic *example* papers, and what the *evidence* says.

Method 1: statistical factors

Advantage

Provides best “**in-sample statistical**” estimates of the factors. Useful for the **APT**, consistent with theory.

Disadvantage

Do not know the **identity** of the factors. Not useful for the ICAPM — no relation to theory. Also subject to **overfitting**: in any small sample, the procedure fits *errors* (noise), not signal/truth.

Example: Roll and Ross (1980), Connor and Korajczyk (1988).

Evidence: **3–5 factors** seem to exist in the U.S. economy; after about five factors, results are insensitive (under either procedure below).

Approach (a): factor analysis

Choose portfolio weights w_j to isolate factor j :

$$\min_{w_j} w_j' \hat{\Sigma} w_j \quad \text{s.t.} \quad w_j' \hat{b}_k = \begin{cases} 0 & \forall k \neq j \\ 1 & k = j \end{cases}$$

— then normalize weights to sum to one.

Method 1: principal components

The first PC is the factor that captures the maximal amount of covariation = the linear combination of assets with the highest possible variance. The second PC is the second most variable portfolio orthogonal to the first.

First principal component

$$\max_{x_1} x_1' \hat{\Omega} x_1 \quad \text{s.t.} \quad x_1' x_1 = 1$$

Solution: the **eigenvector associated with the largest eigenvalue** of $\hat{\Omega}$.

Rescale to form a portfolio:

$$w_1 = \frac{x_1}{\mathbf{1}' x_1}$$

Second principal component

$$\max_{x_2} x_2' \hat{\Omega} x_2 \quad \text{s.t.} \quad \begin{aligned} x_2' x_2 &= 1 \\ x_1' x_2 &= 0 \end{aligned}$$

Solution: the **eigenvector associated with the second largest eigenvalue** of $\hat{\Omega}$.

Rescale to form a portfolio:

$$w_2 = \frac{x_2}{\mathbf{1}' x_2}$$

Under either procedure — factor analysis or principal components — after about **five factors** the results are insensitive.

Method 2: macroeconomic variables

Advantage

Intuitive interpretation of the factors. Consistent with the **ICAPM** — proxies for the changing investment opportunity set or consumption states.

Disadvantage

Hard to measure *unanticipated* changes in macro variables. May not be consistent with the APT — no link to the covariance matrix.

Example: Chen, Roll, and Ross (1986); Chan, Chen, and Hsieh (1985).

Evidence: five factors seem to explain returns best

- (a) growth in industrial production
- (b) changes in expected inflation
- (c) unexpected inflation
- (d) unexpected changes in the difference between returns on corporate and government bonds (default premium)
- (e) unexpected changes in the difference between returns on long- and short-term government bonds (term premium)

Aggregate consumption and oil prices have **no additional effect** beyond these five.

Why use these variables? And how do we measure them? (What about changes in the world's political environment — e.g. Russia. How can we measure this?)

Implementation: the two-step procedure

Step 1: time-series regression to find the betas

Regress security returns on the factors to find each security's **sensitivity to each factor** — for many securities, one security at a time:

$$\tilde{r}_{i,t} - r_{f,t} = \alpha_i + \beta_{i,1}(\tilde{F}_{1,t}) + \cdots + \beta_{i,k}(\tilde{F}_{k,t}) + \tilde{\epsilon}_{i,t}$$

Here the computer is estimating the α 's and β 's for us.

Implementation: the two-step procedure

Step 1: time-series regression to find the betas

Regress security returns on the factors to find each security's **sensitivity to each factor** — for many securities, one security at a time:

$$\tilde{r}_{i,t} - r_{f,t} = \alpha_i + \beta_{i,1}(\tilde{F}_{1,t}) + \cdots + \beta_{i,k}(\tilde{F}_{k,t}) + \tilde{\epsilon}_{i,t}$$

Here the computer is estimating the α 's and β 's for us.

Step 2: cross-sectional regression to find the risk premia

Now regress **average returns** on the betas to see if there is a link:

$$\bar{r}_i - r_f = \hat{\beta}_{i,1}\lambda_1 + \cdots + \hat{\beta}_{i,k}\lambda_k$$

The $\hat{\beta}_i$ are known from Step 1; the slope coefficients are the λ 's, i.e. the **risk premia**.

Implementation: the two-step procedure

Step 1: time-series regression to find the betas

Regress security returns on the factors to find each security's **sensitivity to each factor** — for many securities, one security at a time:

$$\tilde{r}_{i,t} - r_{f,t} = \alpha_i + \beta_{i,1}(\tilde{F}_{1,t}) + \cdots + \beta_{i,k}(\tilde{F}_{k,t}) + \tilde{\epsilon}_{i,t}$$

Here the computer is estimating the α 's and β 's for us.

Step 2: cross-sectional regression to find the risk premia

Now regress **average returns** on the betas to see if there is a link:

$$\bar{r}_i - r_f = \hat{\beta}_{i,1}\lambda_1 + \cdots + \hat{\beta}_{i,k}\lambda_k$$

The $\hat{\beta}_i$ are known from Step 1; the slope coefficients are the λ 's, i.e. the **risk premia**.

Chen, Roll, and Ross (1986): Fama–MacBeth on macro factors

Estimated factor loadings using the past five years of data on **size-sorted portfolios** (portfolios to reduce the errors-in-variables problem; size to maximize dispersion in mean returns). Ran cross-sectional regressions of returns on estimated betas over each of the next 12 months, then a **time-series mean test** on whether the average coefficient differs from zero — exactly the Fama–MacBeth procedure from Lecture 2.

Chen, Roll, and Ross (1986): estimated risk premia

Fama–MacBeth premia on the macro factors; t-statistics in parentheses.

Years	VWNY	MP	DEI	UI	UPR	UTS	Constant
1958–84	−2.403 (−.633)	11.756 (3.054)	−.123 (−1.600)	−.795 (−2.376)	8.274 (2.972)	−5.905 (−1.879)	10.713 (2.755)
1958–67	1.359 (.277)	12.394 (1.789)	.005 (.064)	−.209 (−.415)	5.204 (1.815)	−.086 (−.040)	9.527 (1.984)
1968–77	−5.269 (−.717)	13.466 (2.038)	−.255 (−3.237)	−1.421 (−3.106)	12.897 (2.955)	−11.708 (−2.299)	8.582 (1.167)
1978–84	−3.683 (−.491)	8.402 (1.432)	−.116 (−.458)	−.739 (−.869)	6.056 (.782)	−5.928 (−.644)	15.452 (1.867)

VWNY = return on the value-weighted NYSE index; MP = monthly growth in industrial production; DEI = change in expected inflation; UI = unanticipated inflation; UPR = unanticipated change in the risk premium (Baa-and-under return − long-term government bond return); UTS = unanticipated change in the term structure (long-term government bond return − T-bill rate).

Source: Chen, Roll, and Ross, “Economic Forces and the Stock Market,” *J. Business* 59 (1986).

Reading the Chen–Roll–Ross numbers

Numbers are **percent per month** $\times 10$.

Example: the industrial production (MP) premium

$$\frac{(11.756)(12)}{10} = 14.1\% \text{ per year}$$

is the annual risk premium on the MP factor (growth rate in industrial production). Compare this to roughly **6.5%** for the risk premium on the market factor in the CAPM. Interpretation?

Including a consumption variable and oil prices adds nothing (their Tables 6 and 7).

Note the market index (VWNY) premium is *insignificant* once the macro factors are included.

A question to sit with

Chen, Roll, and Ross motivated their study under the **APT**. Does this make sense? (The APT prices factors that drive the *covariance matrix* — macro series need not.)

Method 3: firm characteristics

Advantage

Intuitive; does not assume constant covariances; uses **returns**, which are unpredictable and therefore better measure *unexpected* changes. Also seems to be **most empirically correct**.

Disadvantage

Data mining concern — or may be picking up *mispricing* (behavioral), which may be unrelated to correlation among securities or consumption states.

Example: Fama and French (1993, 1996).

Evidence: the Fama–French three-factor model

- (a) A portfolio long the market and short T-bills ($Mkt - r_f$)
- (b) A portfolio long small stocks, short large stocks (SMB : “small minus big”)
- (c) A portfolio long high book-to-market stocks, short low book-to-market stocks (HML : “high minus low”)

Risk premia are simply the **average returns on the factor-mimicking portfolios**; Fama–French (1993) illustrate the procedure (see Lecture 2).

Another question to sit with

Fama and French (1993) motivate their study under the **ICAPM**. Does *this* make sense?

The Anomalies in Detail

The size effect

Average annualized returns, beta, and firm size for value-weighted size portfolios of NYSE and AMEX stocks, Apr. 1951 – Dec. 1989 (Hawawini and Keim, 1995).

Size portfolio	Annualized mean return (%)	Beta	Market cap (\$ millions)
Smallest	19.8	1.17	9.7
2	17.8	1.19	23.2
3	16.1	1.15	41.4
4	15.4	1.17	68.0
5	16.0	1.11	109.8
6	14.5	1.05	178.9
7	14.4	1.04	291.4
8	14.8	1.03	502.3
9	13.0	1.01	902.1
Largest	11.9	0.95	3983.0

Small stocks have higher standard deviations and betas ...

... but *not high enough* to explain their returns.

Two readings

- ⇒ small stocks may be **mispriced**,
- ⇒ or small stocks may be correlated with a **missing element of risk**.

The size effect, updated through June 2026

Value-weighted size deciles, NYSE breakpoints, all NYSE/AMEX/NASDAQ stocks. Apr. 1951 – Jun. 2026.

Size portfolio	Annualized mean (%)			Beta 51–26	Avg. cap (\$M, 6/26)
	51–89	90–26	51–26		
Smallest	14.3	12.3	13.3	1.08	206
2	14.1	12.8	13.5	1.17	881
3	15.0	13.2	14.1	1.17	1,849
4	14.8	12.2	13.5	1.15	3,199
5	14.6	12.5	13.6	1.14	4,394
6	13.9	12.4	13.2	1.10	7,389
7	13.7	12.6	13.2	1.10	9,958
8	13.4	12.6	13.0	1.07	20,235
9	12.7	12.2	12.4	1.00	39,437
Largest	11.6	11.5	11.6	0.95	321,092
<i>Small – Large</i>	<i>2.7</i>	<i>0.8</i>	<i>1.7</i>		

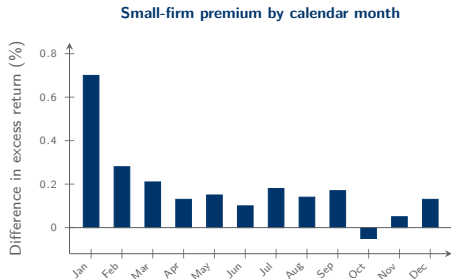
- The small-minus-large spread has **all but closed**: 2.7 points a year through 1989, 0.8 **since 1990**.
- Since 1990 the smallest decile earns *less* than **six of the nine** larger deciles — the monotone pattern is gone.
- Small still carries the higher beta (1.08 vs 0.95), so **risk-adjusted** there is almost nothing left.

Out of sample

The size effect was documented in the early 1980s. Much of it **did not survive** the decades that followed — the classic argument for out-of-sample testing.

Computed from Kenneth R. French's data library (Portfolios Formed on Size; Fama/French 3 Factors), July 2026 CRSP database. Betas from a market-model regression of excess decile returns on the excess market return, Apr. 1951–Jun. 2026. All three columns use French's construction, which differs from Hawawini–Keim's NYSE/AMEX-only deciles — so levels are not directly comparable to the previous slide.

The January effect



The pattern

Much of the small-firm effect seems to occur **in January** — in fact, *3/4 of the small-firm premium occurs in the first five days of January!*

Candidate explanations

Tax-loss selling — but the effect is widespread internationally, even with no capital gains tax; and a size effect remains after controlling for it.

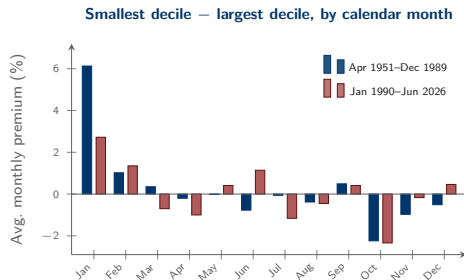
'Window dressing' by institutional investors.

Infusion of capital at the beginning of the year.

Efficiency?

May not accord with efficient markets — why don't people buy in December in anticipation?

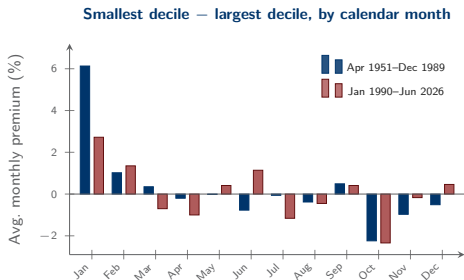
The January effect, updated through June 2026



- January is **still** far and away the biggest month — the effect did *not* vanish.
- But it has **more than halved**: 6.1% a January before 1990, 2.7% since.
- January now accounts for **more** of the annual premium than before ($4.1\times$ the annual total, vs $2.2\times$) — because the other eleven months got *worse*, not because January got better.
- October remains reliably **negative** in both eras.

Computed from Kenneth R. French's data library (Portfolios Formed on Size, value-weighted), July 2026 CRSP database. Premium = smallest size decile minus largest size decile, average monthly return within each calendar month.

The January effect, updated through June 2026



- January is **still** far and away the biggest month — the effect did *not* vanish.
- But it has **more than halved**: 6.1% a January before 1990, 2.7% since.
- January now accounts for **more** of the annual premium than before ($4.1\times$ the annual total, vs $2.2\times$) — because the other eleven months got *worse*, not because January got better.
- October remains reliably **negative** in both eras.

The efficiency question, sharpened

Thirty-five years after the anomaly was published and widely traded, a 2.7% January premium survives.
Decay, but not arbitrage away.

Computed from Kenneth R. French's data library (Portfolios Formed on Size, value-weighted), July 2026 CRSP database. Premium = smallest size decile minus largest size decile, average monthly return within each calendar month.

Other seasonal patterns

Day-of-the-week effect

Returns are on average **negative** on Monday, and on average **positive** Wednesday through Friday.

Weekend effect

From Friday close to Monday, the return drops dramatically. Volume and volatility are *higher on Friday than Monday* — despite the fact that information released over the weekend cannot be traded on until the market opens.

Turn-of-the-month and holiday effects

Returns are much larger in the days preceding market closures for holidays.

Lakonishok and Smidt (1988) provide an excellent review of these anomalies — and address the **data mining critique**.

Contrarian strategies: losers and winners

DeBondt and Thaler (1985, 1987).

The strategy

Take a **three-year** period and rank stocks on their performance over that period.

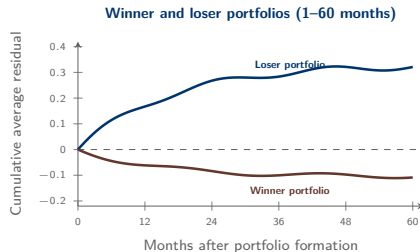
Form a **“winner”** portfolio of the top 10% best performers.

Form a **“loser”** portfolio of the bottom 10% worst performers.

This is called a **contrarian strategy**: look at returns over the *next* few years.

The finding

The **loser portfolio outperforms the winner portfolio** over the subsequent 1–5 years.



Cumulative average residuals for winner and loser portfolios of 35 stocks, 1–60 months into the test period (after DeBondt and Thaler 1985, Figure 3).

Two explanations, and how to account for risk

Overreaction (behavioral)

The winners are firms people have become *too* excited about. Subsequently they realize they were too optimistic — the price falls and returns are low.

Risk (rational)

The losers are *riskier* firms — their higher returns are just compensation for risk.

How do we account for risk?

Variance; or use regression analysis and a pricing model (CAPM).

Examine *when* the strategy exhibits its highest and lowest payoffs — does it do well when the market does? at business-cycle peaks? in recessions?

Catastrophe risk.

From past returns to better forecasting variables

Variation through time in returns is only *partly* due to variation in expected returns, so past returns are a **noisy proxy** for expected returns and the tests lack power. Researchers therefore moved to other, less noisy forecasting variables — D/P , E/P . Is this consistent with efficient or inefficient markets?

Value and growth

The strategy

Form portfolios of **value** stocks — low price relative to fundamentals, i.e. *high* book-to-market (B/M), cash-flow-to-price (C/P), or earnings-to-price (E/P) ratios. Also form portfolios of **growth** stocks — low values of these ratios.

The finding

Value stocks **dramatically outperform** growth stocks. Why?

Rational reading

Represents a **distress factor** in the economy. Value stocks are more prone to this source of risk than growth stocks $\Rightarrow$ higher average returns. Value stocks are typically '*fallen angels*'.

Irrational reading

Growth stocks are '**glamorous**'. People stampede toward them, pushing up the price and depressing future returns; value stocks have been *neglected*, causing prices to fall and expected returns to rise. This is an **overreaction** story.

The strategy

Form portfolios of stocks that performed well **on a relative basis** in the *recent* past ('winners') — i.e. over the past 3 months to one year.

Similarly, form a portfolio of 'loser' stocks.

The finding

The **winners outperform the losers** — in sharp contrast to the 3–5 year mean-reversion (contrarian) result.

Source of the profit?

Underreaction to information? Or is it **risk**?

The Efficiency Debate and Limits to Arbitrage

The case for inefficiency: Shleifer's three arguments

To show markets are inefficient, need to show (i) people make errors in setting prices, and (ii) arbitrage fails to eliminate these errors.

Shleifer (2000): three classical arguments *for* market efficiency

1. Investors are rational and value securities rationally.
2. If some investors are irrational, they are *randomly* so — their trades cancel out.
3. If irrational investors behave similarly, a few rational arbitrageurs can eliminate their influence on prices; irrational investors lose money on average and are driven out of the market.

The case for inefficiency: Shleifer's three arguments

To show markets are inefficient, need to show (i) people make errors in setting prices, and (ii) arbitrage fails to eliminate these errors.

Shleifer (2000): three classical arguments *for* market efficiency

1. Investors are rational and value securities rationally.
2. If some investors are irrational, they are *randomly* so — their trades cancel out.
3. If irrational investors behave similarly, a few rational arbitrageurs can eliminate their influence on prices; irrational investors lose money on average and are driven out of the market.

... and his three rebuttals

1. We can show people behave **less than rationally** (psychological experiments, anecdotal evidence).

The case for inefficiency: Shleifer's three arguments

To show markets are inefficient, need to show (i) people make errors in setting prices, and (ii) arbitrage fails to eliminate these errors.

Shleifer (2000): three classical arguments *for* market efficiency

1. Investors are rational and value securities rationally.
2. If some investors are irrational, they are *randomly* so — their trades cancel out.
3. If irrational investors behave similarly, a few rational arbitrageurs can eliminate their influence on prices; irrational investors lose money on average and are driven out of the market.

... and his three rebuttals

1. We can show people behave **less than rationally** (psychological experiments, anecdotal evidence).
2. People make **similar (correlated) errors** in decision making — framing, loss aversion, overreaction, underreaction.

The case for inefficiency: Shleifer's three arguments

To show markets are inefficient, need to show (i) people make errors in setting prices, and (ii) arbitrage fails to eliminate these errors.

Shleifer (2000): three classical arguments *for* market efficiency

1. Investors are rational and value securities rationally.
2. If some investors are irrational, they are *randomly* so — their trades cancel out.
3. If irrational investors behave similarly, a few rational arbitrageurs can eliminate their influence on prices; irrational investors lose money on average and are driven out of the market.

... and his three rebuttals

1. We can show people behave **less than rationally** (psychological experiments, anecdotal evidence).
2. People make **similar (correlated) errors** in decision making — framing, loss aversion, overreaction, underreaction.
3. In practice, **arbitrage is limited**.

If anomalies are inefficiencies, why doesn't arbitrage eliminate them?

- A **close substitute** for the securities is often not available — so the “arbitrage” is risky. (What types of markets are likely to be more efficient / have more arbitrageurs, and why?)
- Sometimes the strategy can do **very poorly**, and you lose a lot of money.
- Money managers and individuals may have **short horizons** (regular evaluations, psychological preferences): a mispricing can take a while to close — it may not close within the investor's horizon → investors restrict position sizes.
- There can be **high transactions and trading costs** due to turnover; liquidity can be low precisely when you need it most; monitoring can be costly.
- There are **short-sales constraints**.

If anomalies are inefficiencies, why doesn't arbitrage eliminate them?

- A **close substitute** for the securities is often not available — so the “arbitrage” is risky. (What types of markets are likely to be more efficient / have more arbitrageurs, and why?)
- Sometimes the strategy can do **very poorly**, and you lose a lot of money.
- Money managers and individuals may have **short horizons** (regular evaluations, psychological preferences): a mispricing can take a while to close — it may not close within the investor's horizon → investors restrict position sizes.
- There can be **high transactions and trading costs** due to turnover; liquidity can be low precisely when you need it most; monitoring can be costly.
- There are **short-sales constraints**.

The Shleifer–Vishny (1997) mechanism

When clients see bad performance, they get nervous and **pull their money out**. So when mispricing becomes *more* severe (implying *higher* expected returns), arbitrageurs may be unable to get funding — and may even have to liquidate. **When arbitrage is needed most, arbitrageurs are most constrained.**

Finally, all of these so-called anomalies may be subject to the data mining critique.

The critique

If you try enough variables, *something* will eventually appear to predict returns.

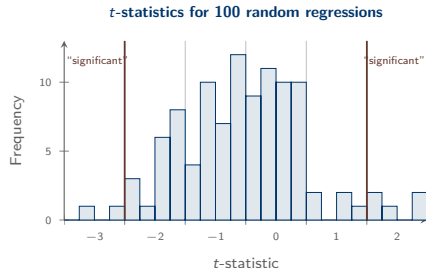
But the forecast power will be completely **spurious** — it won't work out of sample.

Thought experiment

Generate **100 series of completely random numbers**; regress actual stock returns on each. Some *will* produce significant t -statistics — by construction, about 5.

Does this mean you can make money?

NO! Remember this when people show you strategies that worked great in the past — they probably won't work in the future.



So, how efficient is the market?

When confronting evidence of predictable return behavior, ask three questions.

1. Rational variation in expected returns?

Could be tied to CAPM β risk — maybe it represents important conditioning information.

Could represent an **omitted risk factor** from the APT — should explain *common variation* in returns.

Could be correlated with **consumption growth** — consumption states investors care about in the ICAPM or consumption-based models.

2. Irrational deviations of price from fundamental value?

Need to tie to **psychological evidence**, a model of investor sentiment.

Need to show these irrational investors *matter* and can influence prices.

Need to show how **arbitrage fails** to eliminate the errors.

3. Or just chance: data mining, sample-specific conditions?

Testing out of sample is the key — use new data.

**Case study: can
the market add and
subtract?**

Lamont–Thaler: mispricing in tech stock carve-outs

Owen Lamont and Richard Thaler (JPE 2003).

The law of one price

Identical securities must have identical prices. **No-arbitrage** is the central concept of modern finance. Equity prices should also be *non-negative* — why?

They study **equity carve-outs** which seem to violate the law of one price — and which seem to present *near-arbitrage* opportunities.

The number of cases is small, but the mispricings are **large**.

Why is the law of one price easier to test than the EMH?

No asset pricing model is needed — two claims on the *same* cash flows must trade at the same price, whatever the right discount rate is. **No joint hypothesis problem**. (Does efficiency imply the law of one price? How about the reverse?)

Example: 3Com and Palm

On 3/2/00, 3Com carried out an equity carve-out IPO for Palm.

The arithmetic on the day of the IPO

Palm closed at **\$95.06** a share.

Based on the relative number of shares, one share of 3Com indirectly owned **1.5 shares of Palm**.

Thus, based on 3Com's ownership of Palm *alone*, 3Com shares were worth at least

$$1.5 \times \$95.06 = \$142.59.$$

Despite this high implied value, 3Com's actual price was **\$81.81**.

The negative stub

The “**stub value**” — the implied value of 3Com's non-Palm assets and businesses — is

$$\$81.81 - \$142.59 = -\$60.78 \text{ per share.}$$

The market valued 3Com's profitable non-Palm businesses at *minus* \$25 billion.

The trading strategy

How to profit

3Com had **publicly announced its intention to spin off** its remaining Palm shares. (Why is this important?)

To profit: **buy one share of 3Com, short 1.5 shares of Palm, and wait less than a year** — buying a security worth at worst zero for $-\$60.78$. If the spin-off occurs, one is guaranteed a profit.

The situation is *not exactly riskless* (why not?), but it comes close.

The **short horizon** makes this less risky, in some sense, than other alleged mispricings. (Why?)

Three questions, and the Lamont–Thaler answers

- (1) Is this really mispricing? — **Yes**. Fairly unambiguous, based on evidence from the options market; no plausible rational explanation.
- (2) What prevents the law of one price from being enforced? — **Short-sale constraints**.
- (3) Why would anyone buy the overpriced security? — Not sure. The authors say: “*We hypothesize that they are morons.*” — but could there be other explanations?

Transactions costs: dysfunctional yet “efficient”?

Fama's (1991) operational definition again

An efficient market is one in which deviations from the extreme version of the efficiency hypothesis are **within information and trading costs**. Thus one could have “mispricing” as long as it is within transactions costs.

A disturbing example: short-sale constraints

Suppose it is *impossible* to short a stock (infinite transactions costs). Then a stock could be **infinitely overpriced**, yet since there is no way for arbitrageurs to make money, the market is still “efficient.” Can an “efficient market” be infinitely far from a well-functioning, frictionless market?

So in the carve-out cases, we are either ...

rejecting the EMH-with-low-transactions-costs hypothesis, or

accepting the EMH, but conceding that transactions costs make the market imperfect and poorly functioning.

The sample: constructing stub values

Sample

18 carve-outs, 1996–2000: every carve-out in which the parent declared its intention to distribute the remaining shares. (Why is this important?)

Define the stub using the **ratio of issuer shares to be given to parent shareholders** at the distribution date.

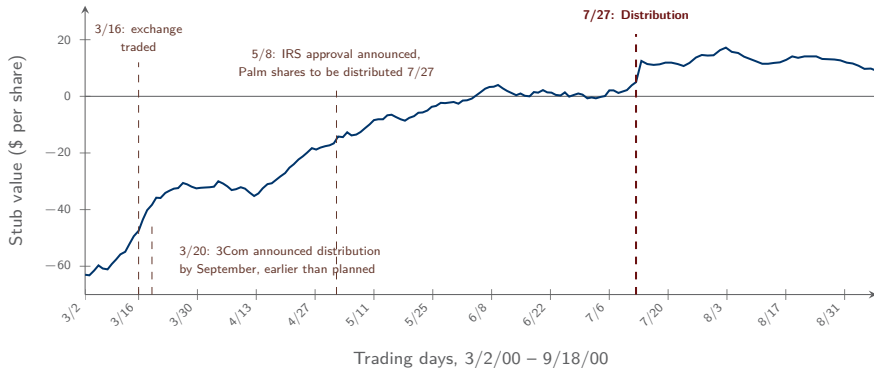
The ratio changes with the parent's share count, so the expected ratio is estimated at each date; the uncertainty is usually small.

Negative stubs: six unambiguous cases, all NASDAQ tech, 12/98–8/00

Parent/Issuer	Min stub (\$ per parent share)	Min stub / Parent	First day negative	Last day negative
Creative/UBID	−74.81	−1.40	1	113
HNC/Retek	−49.01	−0.56	2	177
Daisytek/PFSWeb	−13.72	−0.63	1	81
Metamor/Xpedior	−5.26	−0.19	3	40
3Com/Palm	−62.68	−0.77	1	47
Methode/Stratos	−20.86	−0.39	1	133

The 3Com/Palm stub through time

3Com/Palm stub, 3/2/00 – 9/18/00 (Lamont–Thaler, Figure 5)



The stub remains deeply negative for weeks after the IPO, converging toward zero only as the distribution approaches — and turning positive once the spin-off is completed.

Risk and return on the stub trades

If the stub goes from negative to positive . . .

. . . it must be that the **parent's return is higher than the issuer's**. Looking at returns from the first day of a negative stub to the day after the (distribution) announcement, parents earn returns **33 percent higher** than issuers — over a horizon of about six months.

Magnitudes

The strategies have risk–return trade-offs **four times more favorable than the market's**. These strategies are *grotesquely mispriced* by the CAPM.

Risks specific to stubs: negative events that could have occurred, but didn't

The IRS might not approve the tax-free spin-off; the parent might change its mind; a third party might take over the parent.

Maybe return estimates are too high and risk estimates too low?

After examining the specifics, it doesn't seem plausible that some big negative event could explain these prices — and the **options-market evidence** (next) is not affected by non-completion risk at all.

Short-sale constraints: how the shorting market (doesn't) work

The mispricings were **not easily exploitable**: frictions in the institutions for shorting stocks prevented traders from trading on these opportunities.

With irrational traders, short-sale constraints can cause some stocks to become **overpriced**.

Hard to borrow

To sell short, one must **borrow the stock**, and for institutional reasons borrowing shares can be difficult or impossible for many equities — an investor needs to find an institution or individual willing to lend.

The price of borrowing

If a stock can be found, there is also a price: the **rebate** on short-sale proceeds, negotiated and varying over stocks and time. The rebate can be *negative* — the short-seller pays a daily fee to the borrower.

In the stub situations

Anecdotal reports of both **negative rebates** and stocks **impossible to borrow** — a highly illiquid market in which rebates do not totally equilibrate supply and demand.

Evidence from options: put–call parity

Put–call parity and its American-option cousin

For European options with no transactions costs, parity holds *exactly*; a weaker condition holds for non-dividend-paying *American* options:

$$C - P = S - PV(X) \qquad C - P > S - X$$

For at-the-money options, the inequality says **call prices should exceed put prices**.

The inequality is massively violated for Palm

Synthetic shares of Palm (constructed from options plus borrowing and lending) should have the same price as actual shares, plus or minus transactions costs. They don't: synthetic longs and shorts can be created (how?) — **another violation of the law of one price**, consistent with the difficulty of shorting Palm.

Implied holding cost of borrowing Palm

An investor with a (at least) two-month horizon should be indifferent between shorting actual Palm shares from March until May and creating a synthetic short. Backing out the implied cost: **14 percent over two months — about 119 percent at an annual rate**. Shorting Palm was incredibly expensive.

Ross (1987): “to make a parrot into a learned financial economist, he only needs to learn the single word *arbitrage*.”

Time to teach the parrot new tricks.

The new issues are **unambiguously overpriced** — cheaper to buy parent shares; cheaper to buy calls and sell puts. No rational reason to hold these issues in equilibrium.

Market prices are wrong — though the mispricing is *not easily exploitable*.

Frictions matter, especially short-sale constraints: the implied cost of shorting is very high.

Limits to arbitrage are key.

Hard to know whether this mispricing is confined to a handful of cases — or reflects a more widespread phenomenon.

But, **what is the behavioral or psychological phenomenon driving these mispricings?**

Takeaways

Market efficiency: summary

The framework

Efficiency = prices fully reflect information, given a model of expected returns. The **joint hypothesis problem** means efficiency is never tested alone. Operationally: are deviations within **information and trading costs**?

The evidence

Short-horizon autocorrelations are small (negative for stocks, positive for portfolios via **cross-autocorrelations**); long-horizon returns show mean reversion but with low power. Anomalies — **size, January and seasonals, contrarian reversals, value, momentum** — improve Sharpe ratios beyond the market's. **We will be exploring the major anomalies found in the finance literature and used for quantitative strategies in practice.**

The interpretations

Every anomaly has three readings: **rational** time-varying expected returns / omitted risk factors (multifactor models: statistical, macro, characteristic-based); **irrational** mispricing sustained by *limits to arbitrage*; or **data mining**. Out-of-sample testing discriminates.

The punchline from Lamont–Thaler

Even a *near-arbitrage* — a negative stub with an announced spin-off — can persist for months when shorting is constrained. **Prices can be wrong without being exploitable.**